SHARJAH ASTRONOMICAL OBSERVATORY (SSAH)

SSAH SST Optical Data Processing Pipeline

Technical Reference Manual

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This manual describes the design, scientific conventions, configuration and validation of a Python pipeline that converts raw optical FITS observations into quality-controlled, star-referenced astrometry and photometry of resident space objects, linked tracklets, TLE correlations, and CCSDS TDM / ADES data products.

Document Control

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Revision history

Version	Date	Changes
1.0	June 2026	Initial internal release (software 1.0.0).
2.0	November 2026	First public release. Complete revision for software 2.0.0. Corrected TLE propagation chain, mid-exposure epochs, segmentation-based trail measurement, kinematic tracklet linking, verified astrometry against a local Gaia DR3 store, CCSDS TDM output, extended QC dimensions, validation chapter. Online plate solving and live TLE retrieval removed (offline operation).

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1 Overview

The pipeline post-processes optical observations of resident space objects (RSOs) taken at the Sharjah Astronomical Observatory. Starting from raw FITS frames it produces:

- a per-frame quality-control (QC) matrix with independent photometric and astrometric decisions;
- calibrated, star-referenced RSO positions at the correct mid-exposure epochs, with uncertainties;
- calibrated magnitudes in the reference-catalogue band;
- tracklets linking each object across frames, correlated with a TLE catalogue, and uncorrelated tracks (UCTs) for objects that match no catalogued orbit;
- exchange files in CCSDS Tracking Data Message (TDM) and IAU ADES format.

1.1 Design principles

Two quality tracks

Astrometry and photometry tolerate different defects. A trailed, partly saturated or low-SNR frame may be useless for brightness but still yield a valid centroid. Every frame receives two independent decisions, `photometry_pass` and `astrometry_pass`.

Verified astrometry only

A WCS written by the mount's pointing model is typically tens of arcseconds to arcminutes in error. Positions are reported only from frames whose plate solution has been verified and refitted against reference stars.

Correct epochs

Every measurement is time-stamped at the mid-exposure of the pixels it was measured on, including clock bias and rolling-shutter delays. For low-Earth-orbit objects, timing is usually the dominant error source.

Offline by design

Reference stars come from a local Gaia DR3 store built in advance and TLEs from a local file. No step depends on an external service at run time.

Configuration, not code

Every threshold and instrument parameter is in one YAML file. A site file contains only the values that differ from the defaults.

Batch safety

A corrupt or anomalous frame is recorded as a QC failure with its reason; it never stops a run.

1.2 Key characteristics

Property	Detail
Processing mode	Post-processing of an observing session, offline
Tracking modes	Sidereal (stars are points, RSOs trail) and target-rate (RSO is a point, stars trail)
Quality control	Twelve criteria, two independent decisions per frame
Astrometric reference	Gaia DR3, proper motions applied to the observation epoch
Epochs	Mid-exposure UTC, time bias and rolling shutter, microsecond output
Known-object correlation	SGP4 with light-time and star-referenced convention; along/cross-track O−C
Exchange formats	CCSDS TDM (KVN, RADEC); ADES PSV
Validation	78 automated tests, end-to-end run on synthetic data with ground truth, SGP4 cross-check against skyfield
Requirements	Python $\geq$ 3.10; astropy, sep, photutils, sgp4, astroscrappy, numpy, scipy, pandas

Table 1: Pipeline key characteristics.

2 Installation and Quick Start

2.1 Installation

```
git clone https://github.com/YOUR-GITHUB-USER/ssah-sst-pipeline
cd ssah-sst-pipeline
pip install -e ".[all]"     # pipeline, online catalogue tools, tests
pytest                      # 78 tests, about 40 s
```

The optional extra `online` installs `astroquery`, needed only to download reference stars. The astrometry.net `solve-field` program is optional and used only for frames without a WCS in the header.

2.2 Command-line tools

Command	Purpose
`sst-pipeline`	Process a folder of FITS frames (Section 3).
`sst-refcat`	Build or extend the local Gaia DR3 reference store (Section 9.1).
`sst-tle`	Download a TLE catalogue from CelesTrak for offline use.
`sst-synthetic`	Generate a synthetic sequence with ground truth.
`sst-demo`	Generate, process and compare against truth in one step.

Table 2: Console commands installed with the package.

2.3 First run

```
sst-demo                      # synthetic end-to-end check

cp config/site_template.yaml config/ssah_site.yaml
#   ... fill in every [REQUIRED] value ...
sst-refcat --geo-belt         # once, on a networked machine
sst-tle --group active --out data/tle_catalog.txt
sst-pipeline /data/night1 /results/night1 \
    --config config/ssah_site.yaml
```

3 Processing Stages

Stages 1–8 run for every frame; stages 9–12 run once per session, because they need all frames.

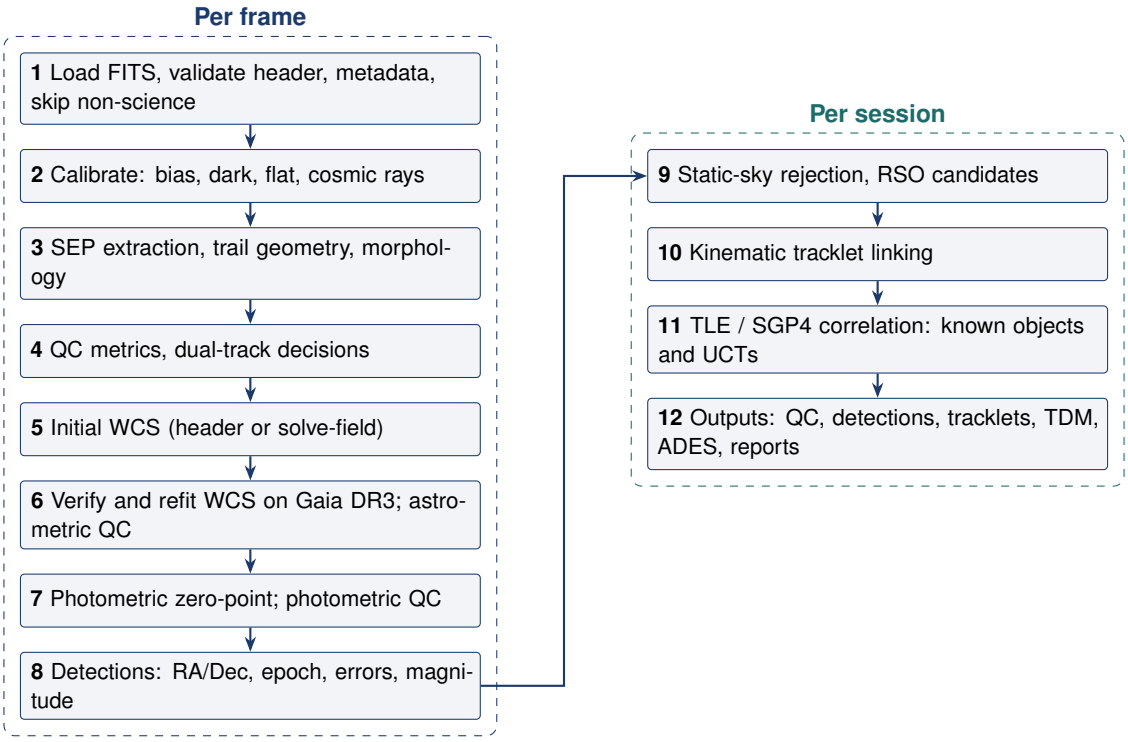


Figure 1: Processing flow. A frame failing the astrometry track after stage 4 or 6 produces no detections.

#	Module	Function
1	`io/fits_loader`, `io/header_validator`, `io/metadata_extractor`	Load frames (incl. compressed and multi-extension), validate headers, extract metadata; skip bias/dark/flat frames by `IMAGETYP`.
2	`calibration/calibration_manager`	Bias, scaled dark and flat correction; L.A.Cosmic cleaning.
3	`detection/source_extractor`, `detection/source_classifier`	One SEP pass; moment and profile trail geometry; fragment merging; POINT / TRAILED / ARTIFACT / EXTENDED.
4	`qc/metrics`, `qc/qc_matrix`	Image-quality metrics and dual-track decisions.
5–6	`astrometry/wcs_solver`, `astrometry/reference_catalog`, `astrometry/catalog_store`	Initial WCS; offset vote, one-to-one match and TAN+SIP refit against Gaia DR3.
7	`photometry/aperture_photometry`	Zero-point from the same reference stars; CCD-equation errors.
8	`astrometry/position_extractor`	Sky position, mid-exposure epoch, uncertainty, magnitude.
9–10	`tracking/linker`	Static-sky test, candidate selection, constant-rate linking.
11	`tracking/tle_manager`, `tracking/orbit_propagator`, `tracking/known_object_matcher`	SGP4 prediction and correlation.
12	`output/writers`, `reporting/report_generator`	Data products, workbook, plots.

Table 3: Stages and modules.

4 Configuration

The packaged file `sst_pipeline/defaults.yaml` holds every parameter with comments. A site file (copied from `config/site_template.yaml`) is deep-merged over it, so it contains only site-specific values. Keys that do not exist in the defaults trigger a warning, so a misspelled parameter is not silently ignored.

Parameter	How to obtain
`instrument.gain_e_per_adu`	Camera datasheet or photon-transfer curve.
`instrument.read_noise_e`	Camera datasheet, or bias-frame statistics.
`instrument.saturation_adu`	The non-linearity limit, not the ADC maximum.
`instrument.pixel_scale_arcsec`	$206265\times$ pixel size / focal length; confirmed by plate solutions.
`site.latitude_deg`, `longitude_deg`, `height_m`	Surveyed geodetic coordinates.
`site.iau_obs_code`	MPC observatory code, if assigned (ADES only).
`observation.tracking_mode`	`sidereal` or `rate`.
`timing_calibration.date_obs_reference`	Whether the camera software writes exposure start, mid or end.
`timing_calibration.time_bias_ms`	Measured on sky (Section 11).
`timing_calibration.shutter_type`, `readout_time_ms`	`global`, or `rolling` with readout time (CMOS).
`calibration.master_*_path`	Master bias (ADU); dark bias-subtracted per second; flat normalised, same filter.
`astrometry.reference_catalog_dir`	Local Gaia DR3 store (Section 9.1).
`tle_tracking.tle_file_path`	TLE file from `sst-tle`; refresh daily for LEO.

Table 4: Site parameters to set before processing real data.

5 FITS Header Validation

Keyword rules are declared in the configuration. A missing or unparsable mandatory keyword fails both QC tracks; optional keywords produce warnings.

Keyword	Aliases	Type and purpose
Mandatory		
DATE-OBS	—	ISO-8601 UTC; date-only values are combined with TIME-OBS / UT. Epoch of every measurement.
EXPTIME	EXPOSURE	0.0001–600 s. Mid-exposure epoch, rates, fluxes.
OBJECT	TARGET	Links the frame to the observing plan.
TELESCOP	TELESCOPE	Instrument traceability.
INSTRUME	INSTRUMENT, CAMERA	Camera identification, calibration matching.
Recommended		
RA, DEC	OBJCTRA, OBJCTDEC	Degrees or sexagesimal strings ("10 00 00.0"). Plate-solve hint.
WCS keywords	CTYPEi, CRVALi, CRPIXi, CD/CDELT	Initial plate solution (verified before use).
IMAGETYP	IMGTYPE, FRAMETYP	Non-science frames (bias, dark, flat) are skipped.
AIRMASS, FILTER, CCD-TEMP, OBSERVER	—	Recorded; airmass is computed from site and pointing when absent.

Table 5: Header keywords.

6 Calibration

Science frames are reduced as

$$I_{\mathrm{cal}} = \frac{I_{\mathrm{raw}} - B - D_{1\,\mathrm{s}}\, t_{\mathrm{exp}}}{F}, \tag{1}$$

where B is the master bias, $D_{1\,\mathrm{s}}$ the bias-subtracted dark per second and F the normalised flat. Masters whose shape differs from the frame are rejected and logged. Cosmic rays are removed with L.A.Cosmic [3] (astroscrappy). The cosmic-ray mask is kept for QC. Saturation and hot pixels are measured on the raw frame, because calibration moves pixels away from the detector ceiling.

7 Source Extraction and Trail Measurement

A single SEP [1, 2] extraction per frame serves QC, plate verification, photometric calibration and target measurement. Each source is re-measured from its segmentation pixels with flux-weighted second moments $\lambda_1 \geq \lambda_2$ along and across its principal axis. For a uniform trail of length L convolved with a PSF of width σ:

$$\lambda_1 = \frac{L^2}{12} + \sigma^2, \qquad \lambda_2 = \sigma^2 \quad \Rightarrow \quad L = \sqrt{12\,(\lambda_1 - \lambda_2)}. \tag{2}$$

Profile refinement. Moments are biased when a trail crosses a star or is split by deblending. The along-track profile within $\pm 2.5\sigma$ of the axis,

$$P(s) = P_0 \left[\Phi\!\left(\tfrac{L/2 - s}{\sigma}\right) - \Phi\!\left(\tfrac{-L/2 - s}{\sigma}\right) \right], \tag{3}$$

falls to exactly half its plateau P_0 at the true end points $s = \pm L/2$. The pipeline re-fits the axis from across-track centroids, median-filters $P(s)$, and takes the half-maximum crossings as end

points. The midpoint is the mid-exposure position, and $P_0 L$ gives a flux that is robust to crossing stars. Collinear fragments of one trail are merged before this step.

Classification. `TRAILED`: elongation ≥ 2.5 and $L \geq 8$ px. `ARTIFACT`: sharper than the PSF (hot pixel, cosmic ray), or saturated and aligned with a detector axis (charge bleeding). `EXTENDED`: FWHM above 15 px. Otherwise `POINT`. The role of each class depends on the tracking mode:

Tracking mode	Stars appear	RSOs appear	Reference stars for WCS / ZP
`sidereal`	POINT	TRAILED	POINT sources
`rate`	TRAILED	POINT	TRAILED sources (midpoints)

Table 6: Morphology roles.

8 Quality Control

Each frame receives two decisions. Thresholds are the defaults in `defaults.yaml`; calibrate them on real SSAH data.

Metric	Measured on	Photometry track	Astrometry track
Header validity	header	fail if invalid	fail if invalid
Signal presence	SEP catalogue	≥ 1 source	≥ 1 source
SNR (unsaturated)	SEP catalogue	≥ 10	≥ 5
Saturated pixels	raw frame	$\leq 0.05\%$	$\leq 2\%$
Seeing (FWHM)	reference stars	$\leq 6''$	$\leq 10''$
Star ellipticity	reference stars (sidereal)	≤ 0.35 (tracking/focus)	—
Background stability	rolling window	level and noise within 4σ	—
Transparency	star count	$\geq 50\%$ of sequence median	—
Artefacts	raw / calibrated	CR $\leq 0.2\%$, hot px, columns, gradient	CR $\leq 1\%$
Trail length	SEP catalogue	≤ 2000 px	—
Plate solution	Gaia DR3 fit	—	verified; RMS $\leq 1.5''$
Zero-point scatter	Gaia DR3 stars	≥ 5 stars; RMS ≤ 0.10 mag	—

Table 7: QC metrics and default thresholds.

Background stability and transparency are judged against a rolling window of the last ten valid frames (at least five), using robust statistics: $|x - \tilde{x}| / \max(1.4826\,\mathrm{MAD},\ 0.02\,\tilde{x}) \leq 4$.

9 Astrometry

9.1 Reference catalogue

Reference stars are Gaia DR3 [6] sources (G < 16 by default) stored locally in cells of about $2^\circ \times 2^\circ$. Each cell is one compressed file with position (ICRS, epoch J2016.0), proper motion and G magnitude. At run time only the cells covering a frame are read, and positions are propagated

to the observation epoch:

$$\alpha(t) = \alpha_0 + \frac{\mu_{\alpha*}}{\cos\delta_0}(t - 2016.0), \qquad \delta(t) = \delta_0 + \mu_\delta(t - 2016.0). \tag{4}$$

The store is built in advance with `sst-refcat`, which is resumable and downloads from VizieR (CDS, then the CfA mirror) with the ESA Gaia archive as fallback, each under a timeout:

```
sst-refcat --geo-belt --dry-run        # preview the cells
sst-refcat --geo-belt                  # GEO band seen from the site
sst-refcat --geo-belt --margin 15      # + inclined GEO objects
sst-refcat --site-visible              # sky above 20 deg altitude
sst-refcat --around 150 20 1.5         # one field
sst-refcat --from-csv gaia_export.csv  # import, no download
```

From SSAH (latitude $25.36°$ N, altitude $> 20°$) the geostationary ring appears near declination $-4°$ because of parallax. At G < 16 the GEO belt needs 719 cells, the belt $\pm 15°$ 2 843 cells, and the whole visible sky 8 985 cells. When `sst-pipeline` runs with `-catalog`, cells missing for an observed field are downloaded once and kept.

9.2 Plate verification and refit

1. The initial WCS comes from the header or, with `-solve-wcs`, from a local astrometry.net [5] solve of the calibrated frame with scale and position hints.
2. Reference stars are projected with the initial WCS. A two-dimensional vote over all star–catalogue pixel offsets (search radius $120''$) absorbs pointing errors.
3. Stars are matched one-to-one within $2''$, and a TAN projection is fitted (with SIP distortion of order 2 when at least 15 stars match), with iterative 3σ clipping.
4. The plate is *verified* when at least 8 stars match. Its RMS,

$$\mathrm{RMS}_{\mathrm{astro}} = \sqrt{\langle(\Delta\alpha\cos\delta)^2 + (\Delta\delta)^2\rangle}, \tag{5}$$

must not exceed $1.5''$.

9.3 Positions and uncertainties

Each detection's centroid uncertainty is its RMS width divided by its SNR along each principal axis, rotated to RA/Dec through the local WCS. It is added in quadrature to the plate RMS of the corresponding axis. Pixel coordinates are 0-based, consistent with astropy.

10 Photometry

Point sources use a circular aperture of radius $1.5\times$FWHM with a sigma-clipped annulus (4–$7\times$FWHM). Trails use a rotated rectangle of length $L + 3$ FWHM and half-width 1.5 FWHM with a rectangular sky annulus, as in trailed-source photometry [9]. Both are corrected to total flux with the Gaussian enclosed-energy fraction ($1 - e^{-r^2/2\sigma^2}$ for a circle, $\mathrm{erf}(h/\sqrt{2}\sigma)$ times the along-track fraction for a trail). If stars inside a trail aperture inflate the flux, the profile flux $P_0 L$ is used instead.

The uncertainty follows the CCD equation [4]:

$$\sigma_F^2 = \frac{F}{g} + n_{\mathrm{ap}}\,\sigma_s^2 + \frac{n_{\mathrm{ap}}^2\,\sigma_s^2}{n_{\mathrm{sky}}} \qquad [\mathrm{ADU}^2], \tag{6}$$

with gain g (e^-/ADU) and empirical sky noise σ_s per pixel.

Instrumental magnitudes are $m_{\mathrm{inst}} = -2.5\log_{10}(F/t_{\mathrm{exp}})$. The frame zero-point is the median of $m_{\mathrm{cat}} - m_{\mathrm{inst}}$ over unsaturated reference stars with SNR ≥ 20, with MAD-based clipping. The photometric RMS is the scatter of the retained residuals. A frame zero-point already includes atmospheric extinction, so $m = m_{\mathrm{inst}} + \mathrm{ZP}$. Only when no zero-point can be measured is the configured fallback used, $m = m_{\mathrm{inst}} + \mathrm{ZP}_0 - kX$. Magnitudes from it are flagged as uncalibrated and are not exported.

11 Timing

The epoch of a measurement at detector row y is

$$t = t_{\text{DATE-OBS}} - \Delta_{\mathrm{ref}} + t_{\mathrm{bias}} + \frac{y}{N}\,T_{\mathrm{readout}} + \frac{t_{\mathrm{exp}}}{2}, \qquad \Delta_{\mathrm{ref}} \in \{0,\ t_{\mathrm{exp}}/2,\ t_{\mathrm{exp}}\}, \tag{7}$$

where Δ_{ref} depends on whether DATE-OBS marks the start, middle or end of the exposure. The row term applies only to rolling shutters (`readout_time_ms` over N rows). For uniform motion the flux-weighted trail centroid lies at the mid-exposure position, so the same epoch applies to point sources and trail midpoints. Epochs are written with microsecond resolution.

A timing error δt displaces an object moving at angular rate ω by

$$\delta\theta = \omega\,\delta t. \tag{8}$$

Object	Rate ω	10 ms error	Half of a 2 s exposure
LEO near zenith (e.g. ISS)	$\approx 1^\circ$/s	$\approx 36''$	$\approx 1^\circ$
GEO, sidereal tracking	$15''$/s	$0.15''$	$15''$

Table 8: Effect of timing errors on position.

> **Calibrating the time bias.** Observe GNSS satellites (GPS, Galileo), whose orbits are known to centimetre level, and process them with a TLE or precise ephemeris. The pipeline reports for each identified tracklet the mean along-track O−C divided by the predicted rate (`along_track_time_offset_s`). Averaged over several satellites, this is the clock and shutter bias to enter as `time_bias_ms`.

12 Tracklets and Uncorrelated Tracks

Static sky. A detection whose sky position recurs within $3''$ in at least three frames is fixed on the sky (a star or galaxy below the catalogue limit) and is rejected.

Candidates. RSO candidates are detections that are not catalogue stars, not static, not artefacts, and have the morphology expected for the tracking mode. In sidereal mode, point sources become candidates only when the static test is possible (at least three frames).

Linking. Pairs of candidates in different frames within 120 s whose implied rate lies between 0.2 and $4000''/\mathrm{s}$ seed a constant-rate model in the tangent plane. For trails, the motion direction and rate must also agree with the trail axis ($\pm 10^\circ$) and the trail rate ($\pm 30\%$). Each seed is grown by adding the nearest candidate within $6''$ of the prediction in every later frame. The longest chain with at least three positions and a fit RMS $\leq 3''$ becomes a tracklet. Unlinked trails are kept as single-frame tracklets; a single trail gives a position and a rate, with direction ambiguous by 180°.

13 Known-Object Identification

13.1 Prediction chain

1. SGP4 [7] with WGS-72 constants propagates every catalogue object at once to the measurement epoch, giving TEME position and velocity.
2. Light-time correction: $\mathbf{r}(t - \rho/c) \approx \mathbf{r} - \mathbf{v}\,\rho/c$ (about 25 m for LEO).
3. TEME is transformed to GCRS (astropy [11]), and the site's GCRS position is subtracted to give the topocentric line of sight.
4. **Star-referenced convention.** The plate solution maps the apparent star field onto aberration-free catalogue positions, removing the annual aberration ($\approx 20''$) that stars show. An Earth-orbiting object shows no annual aberration, so its measured position carries the opposite shift. The line-of-sight *direction* is therefore transformed from GCRS to ICRS as a unit vector, which applies the same correction as the plate solution. Diurnal aberration affects stars and satellite alike and cancels.
5. Altitude and azimuth are computed from the Earth-fixed topocentric vector. The apparent rate and direction of motion come from predictions at $t \pm 0.5$ s.

13.2 Matching

TLE errors are dominated by the along-track component [8]. The O−C offset is therefore decomposed along and across the predicted direction of motion and tested against separate tolerances.

Criterion	Default
Along-track |O−C|	$\leq 900''$
Cross-track |O−C|	$\leq 180''$
Trail rate vs predicted rate	within $\pm 30\%$
Trail axis vs predicted direction	within 15° (modulo 180°)
Object altitude	above the horizon
Quality `GOOD`	inside the half-size tolerance ellipse and TLE age ≤ 14 days; otherwise `MARGINAL`

Table 9: Correlation criteria.

A tracklet takes the NORAD identity held by the majority of its detections, provided its fitted rate agrees with the prediction. Tracklets without an identity are reported as uncorrelated tracks (UCTs).

14 Output Files

File	Contents
`qc_matrix.csv`	Per-frame metrics, WCS and zero-point statistics, both decisions with reasons.
`rejected_frames.log`	Human-readable rejection reasons per track.
`session_metadata.csv`	Parsed header values for every science frame.
`skipped_frames.csv`	Non-science frames that were skipped.
`detections.csv`	Every non-star detection: epoch, RA/Dec and uncertainties, magnitude, morphology, status (`LINKED`, `SINGLE_STREAK`, `UNLINKED`, `STATIC`), tracklet, NORAD ID, O−C.
`tracklets.csv`	One row per tracklet: rate, position angle, fit RMS, mean magnitude, identification, along/cross-track O−C, time offset.
`uncorrelated_tracks.csv`	Tracklets not matched to the TLE catalogue.
`observations_tdm.kvn`	CCSDS TDM [10]: `ANGLE_TYPE = RADEC`, `REFERENCE_FRAME = ICRF`, UTC, one segment per tracklet, verified plates only.
`observations_ades.psv`	ADES PSV for archiving. The MPC does not accept artificial-satellite observations.
`qc_summary.json`	Session totals.
`SST_QC_Report.xlsx`	Summary, QC Matrix, Metadata, Tracklets, Detections sheets.
`plot_01–plot_05`	Pass rates, SNR, background, seeing and plate RMS, star counts.
`pipeline_<UTC>.log`	Full session log.

Table 10: Data products written per session.

15 Validation

15.1 Synthetic sequence with ground truth

`sst-synthetic` renders a physically consistent sequence: twelve 1024^2 frames at $0.82''$/px, 2 s exposures every 10 s, Poisson and read noise, and a fixed star field (catalogued and faint uncatalogued stars). It contains three moving objects and quality-control test cases: cosmic rays, cloud, a saturated star with charge bleeding, a missing header keyword, and a bias frame. The header WCS is deliberately offset by $29''$ and rotated by $0.05°$.

	Trailed RSO	Slow point RSO	Fast single streak
Detected as	10-frame tracklet	11-frame tracklet	single streak
Position error, median / max	$0.06''$ / $0.19''$	$0.01''$ / $0.04''$	$0.14''$
Rate, recovered (true)	$6.001''$/s (6.0)	$0.600''$/s (0.6)	$149.99''$/s (150)
Position angle, recovered (true)	$59.96°$ (60)	$200.02°$ (200)	$115.00°$ (115)
Magnitude, recovered (true)	12.51 (12.5)	13.00 (13.0)	10.50 (10.5)

Table 11: Recovery of the synthetic objects.

Plates were corrected by $29.2''$ to an RMS of $0.04''$. Zero-points agree with the injected value to within 0.03 mag. Epochs equal the true mid-exposure times. No star was reported as an RSO. The quality-control outcomes were as designed: the bias frame was skipped, the frame with a

missing keyword failed both tracks, the cloud and bleeding-star frames failed photometry only, and the cosmic-ray frame was cleaned and passed.

15.2 Orbit prediction

For an ISS pass at $43.6°$ altitude, the geometric topocentric predictions agree with the independent skyfield implementation to $1.4″$ in position, $0.0001°$ in altitude and 1 m in range. A known along-track timing offset of 50 ms is recovered by the time-offset estimate to within 2 ms.

15.3 Test suite

The 78 automated tests (`pytest`) cover timing, header parsing, calibration, extraction and trail geometry (including trails crossing bright stars), photometric flux recovery and error model, plate refit, catalogue-store geometry and download fallbacks (with simulated servers), QC decisions, linking, SGP4 correlation and the end-to-end run above.

> This is a validation of logic and accuracy on synthetic data. Thresholds, the time bias, and photometric performance must still be established on real SSAH observations.

16 Limitations and Next Steps

- **Instrument and site values** in the configuration are provisional until confirmed (gain, read noise, non-linearity limit, surveyed coordinates, DATE-OBS convention).
- **Time bias** must be measured on sky before reporting LEO positions.
- **Thresholds** are defaults and should be tuned on real noise statistics.
- **Target-rate tracking** is unit-tested; an end-to-end check on real frames is pending.
- **Live downloads** (VizieR, ESA, CelesTrak) have been exercised against simulated servers only; a first single-field build confirms the path (`sst-refcat -around`).
- **Photometric band**: unfiltered frames are calibrated to Gaia G without a colour term; expect object-colour-dependent offsets of a few tenths of a magnitude.
- **Orbit determination**, conjunction assessment and re-entry analysis are outside the scope. A single short tracklet constrains an orbit poorly; initial orbit determination needs several tracklets or a priori orbits.

17 Acknowledgements, License and Disclaimer

Acknowledgements. The pipeline was developed by Javed Iqbal during an internship at the Sharjah Astronomical Observatory (SSAH). It uses data and services from:

- the European Space Agency mission Gaia, with data processed by the Gaia Data Processing and Analysis Consortium (DPAC);
- the VizieR catalogue service (CDS, Strasbourg);
- CelesTrak (TLE distribution);
- astrometry.net, optionally.

Publications based on Gaia data should include the acknowledgement that ESA requests, given at https://www.cosmos.esa.int/web/gaia-users/credits.

License. The software and this manual are released under the BSD 3-Clause License; see the LICENSE file. Third-party packages are used under their own licenses and are not bundled. Most are BSD- or MIT-licensed; SEP is LGPL-3.0-or-later.

Citing. Please cite the software using the metadata in CITATION.cff, which GitHub offers through its "Cite this repository" button.

Disclaimer. The software is provided "as is", without warranty of any kind. It has been validated on synthetic data. It is not a certified system for collision avoidance or other safety-critical decisions.

A Formula Reference

Quantity	Expression
Measurement epoch	$t = t_{\text{DATE-OBS}} - \Delta_{\text{ref}} + t_{\text{bias}} + (y/N)\,T_{\text{readout}} + t_{\text{exp}}/2$
Position error from timing	$\delta\theta = \omega\,\delta t$
Pixel scale	$206265 \cdot p_{\text{mm}}/f_{\text{mm}}$ [″/px]
Trail length (moments)	$L = \sqrt{12(\lambda_1 - \lambda_2)}$
Trail rate	$\omega = L \cdot s_{\text{pix}}/t_{\text{exp}}$
Astrometric RMS	$\sqrt{\langle(\Delta\alpha\cos\delta)^2 + (\Delta\delta)^2\rangle}$
Proper motion	$\alpha = \alpha_0 + \mu_{\alpha*}\Delta t/\cos\delta_0,\ \ \delta = \delta_0 + \mu_\delta \Delta t$
Flux uncertainty	$\sigma_F^2 = F/g + n_{\text{ap}}\sigma_s^2 + n_{\text{ap}}^2\sigma_s^2/n_{\text{sky}}$
Instrumental magnitude	$m_{\text{inst}} = -2.5\log_{10}(F/t_{\text{exp}})$
Calibrated magnitude	$m = m_{\text{inst}} + \text{ZP}$ (frame ZP); $m = m_{\text{inst}} + \text{ZP}_0 - kX$ (fallback)
Magnitude uncertainty	$\sigma_m \approx 1.0857/\text{SNR}$
Airmass (Rozenberg 1966)	$X = 1/(\cos z + 0.025\,e^{-11\cos z})$
Light-time	$\mathbf{r}(t - \rho/c) \approx \mathbf{r} - \mathbf{v}\rho/c$
Topocentric vector	$\boldsymbol{\rho} = \mathbf{r}_{\text{sat}}^{\text{GCRS}} - \mathbf{r}_{\text{site}}^{\text{GCRS}}$
O−C decomposition	along $= \Delta \cdot \hat{\mathbf{u}}$, cross $= \Delta \cdot \hat{\mathbf{n}}$ (tangent plane)
Time offset	$\Delta t = \langle\text{along}\rangle/\omega_{\text{pred}}$
Angular separation	Vincenty formula (exact at all separations)

Table 12: Formulae used in the pipeline.

B Command Reference

```
sst-pipeline FITS_DIR OUT_DIR
    [--config SITE.yaml] [--catalog] [--solve-wcs]
    [--pattern "*.fits"] [--label TEXT]

sst-refcat REGION [--out DIR] [--mag-limit 16] [--min-alt 20]
    [--margin 2] [--source auto|vizier|esa] [--workers 3]
    [--timeout S] [--retries 2] [--dry-run]
  REGION: --geo-belt | --site-visible | --box RA0 RA1 DEC0 DEC1
          | --around RA DEC RADIUS | --from-csv FILE

sst-tle [--group active | --catnr NORAD] [--out data/tle_catalog.txt]
sst-synthetic [OUT_DIR] [--seed N]
sst-demo [--dir sst_demo]
```

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